Estimation:

In most statistical studies, population parameters are either infeasible to obtain or unknown. Estimation is concerned with the estimation of population parameter from the sample statistic

Estimators

Let 'θ' be any population ~~parame~~ parameter which is to be estimated. Let $x_1, x_2 \ldots x_n$ be a random sample. Then any statistic computed from a random sample to infer value of a population parameter are called point estimator or simply estimator.

The value of the population parameter is called estimate.

Unbiased estimator

Let M be an estimator of the population parameter θ. Then M is said to be unbiased if $E(M) = \theta$

Example: Let $x_1, x_2 \ldots x_n$ be a random sample. Then an unbiased estimate for the mean μ of the population is the sample mean $\bar{X} = \frac{x_1 + x_2 - - x_n}{n}$

since $E(\bar{X}) = \mu$.

Unbiased Estimates for mean and variance (Population mean and variance)

If we take a random samples with values $x_1, x_2 \ldots, x_n$ from a population, then an unbiased estimate for the population mean is the sample mean $\bar{x} = \frac{x_1 + x_2 + \cdots x_n}{n} = \frac{\sum x_i}{n}$

Now we think about an unbiased estimator for the population variance σ^2.

A natural choice would be the sample variance S^2, which is given by $S^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$

But $E(S^2) \neq \sigma^2$ in general (It can be proved)

So S^2 is not an unbiased estimator for the population variance.

However, if we replace the denominator by $(n-1)$, it would be an unbiased estimator for σ^2

and we denote it by $\bar{S}^2$

ie; $\bar{S}^2 = \frac{1}{n-1}\sum(x_i-\bar{x})^2$ is an unbiased estimate for population variance σ^2.

Therefore, If we take a random sample with values $x_1, x_2, \dots x_n$ from a population, then

(1) an unbiased estimate for the population mean μ is the sample mean $\bar{x} = \frac{x_1+x_2+\dots x_n}{n} = \frac{\sum x_i}{n}$

And (2) an unbiased estimate for the population variance σ^2 is the modified sample variance

$$\bar{S}^2 = \frac{1}{n-1}\sum(x_i-\bar{x})^2$$

NOTE

The common estimate S^2 and the unbiased estimate $\bar{S}^2$ of population variance are connected by the relation $\bar{S}^2 = \left(\frac{n}{n-1}\right)S^2$

~~Standard Error (SE)~~

A good estimator should possess the following properties

(1) unbiased (we discussed above)

If the mean value of the sampling distribution of the statistica is equal to the parameter estimated, $[E(m) = \theta]$ then the estimator is unbiased

(2) Efficient: If the estimator has a relatively small variance, then it is efficient

(3) Sufficient: If the estimator use all the information available from the sample, then it is sufficient

(4) Consistent: If the estimator approaches the parameter as the sample size becomes larger, then it is consistent

~~Approximately 60% of graduate stud~~

Standard Error

The standard Error (SE) Of a statistic is the Standard deviation Of its sampling distribution.

The reciprocal of SE of a statistic gives a measure of the precision or reliability of the estimate of the parameter.

We have seen that the sampling distribution of mean $\bar{X}$ can be approximated by a normal distribution $N(\mu, \sigma/\sqrt{n})$ for large n, where μ and σ are the population mean and standard deviation.

1) So the standard error of mean is $\sigma/\sqrt{n}$ when 'n' large

2) If σ is unknown and n is very large, we can replace σ by the unbiased estimate $\bar{S}$, the sample standard deviation. So $SE_{\text{of mean}} = \bar{S}/\sqrt{n}$ when 'σ' is not known and n very large

The sampling distribution of proportion $\hat{p}$ can be approximated by a normal distribution $N(P, \sqrt{PQ/n})$ where P is the population proportion and $Q = 1-P$ when n is large. So

3) The standard error of ~~mea~~ proportion is $\sqrt{PQ/n}$

4) If P is not known, we can replace the unbiased estimate $\hat{p}$ computed from a sample for the population proportion:

$\therefore$ SE of proportion $= \sqrt{\frac{\hat{p}\hat{q}}{n}}$ where $\hat{q} = 1-\hat{p}$ when 'n' is large.

Confidence Interval and Confidence level

A confidence interval for an unknown population parameter is an interval, that is predicted to contain the population parameter with a certain probability called the confidence level

i.e. The probability of an estimator that lie ~~between~~ in an interval is called **confidence level**. That interval is called **confidence interval**.

Usually, the confidence level is chosen by the investigator as a high value expressed in percentages, such as 90%, 95% or 99%.

Thus a 95% confidence level of a parameter θ lie in an interval (c_1, c_2) means that $P[c_1 < \theta < c_2] = 0.95$

Eg: The confidence interval for the population mean

We have the sampling distribution of mean $\bar{X}$ can be approximated by a normal distribution $N(\mu, \sigma/\sqrt{n})$

or $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$ ie, $\bar{X} \sim N(\mu, SE)$ if n is large

$\therefore Z = \frac{\bar{X} - \mu}{SE} \sim N(0,1)$ as n is large (ie, $n \geq 30$)

Suppose, we want to find the interval estimation of μ with confidence level 95%

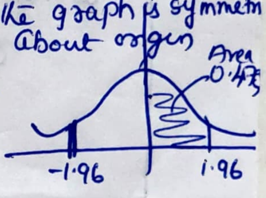
[we usually denote confidence level by $1-\alpha$ and α is called the significance level]

Thus we have to find two constants c_1 and c_2 such that

$P[c_1 < Z < c_2] = 0.95$

$P(0 < Z < c) = 0.95/2 = 0.475$

Since $Z \sim N(0,1)$ the graph is symmetric about origin



From the normal table; $P(0 < Z < 1.96) = 0.475$

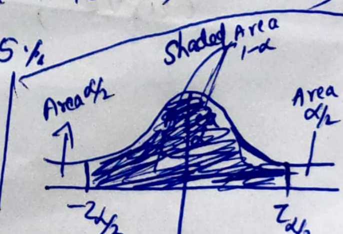
$\therefore P(-1.96 < Z < 1.96) = 0.95$

ie, $P\left(-1.96 < \frac{\bar{X} - \mu}{SE} < 1.96\right) = 0.95$

$\Rightarrow P(\bar{X} - 1.96 \times SE < \mu < \bar{X} + 1.96 \times SE) = 0.95$

$\therefore$ The confidence limits of μ are $\bar{X} \pm 1.96 \times SE$

and the confidence interval is $(\bar{X} - 1.96\,SE, \bar{X} + 1.96\,SE)$ provided if the confidence level is 95%.



Now for any confidence level '$1-\alpha$' we can take the confidence limits of μ as $\bar{X} \pm Z_{\alpha/2} \times SE$ where $Z_{\alpha/2}$ is that value for which $P[-Z_{\alpha/2} < Z < Z_{\alpha/2}] = 1-\alpha$

This $Z_{\alpha/2}$ is called critical value or significant value.

The following table gives the critical values of different confidence levels

Confidence level $(1-\alpha)\%$	90%	95%	99%
$Z_{\alpha/2}$	1.645	1.96	2.58

Confidence interval of mean for large sample

For a large sample with $(1-\alpha)\%$ confidence level, the confidence interval of population mean is given by

$$\left(\bar{x} - Z_{\alpha/2}\, SE,\ \bar{x} + Z_{\alpha/2}\, SE\right) \quad \text{or} \quad \bar{x} \pm Z_{\alpha/2}\, SE$$

where the standard error

$SE = \sigma/\sqrt{n}$ when the population standard deviation σ is known.

and

$SE = \bar{s}/n$ when the population standard deviation σ is unknown where $\bar{s}$ is the ~~unbiased~~ sample standard deviation.

ie, $\bar{s}^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$ (If $\bar{s}$ is not given use this formula)

Here the critical values $Z_{\alpha/2}$ are the Z-scores which ~~cutt~~ cut off an area of $\alpha/2$ on both the tails of the ~~standaard~~ standard normal curve, so that $P(-Z_{\alpha/2} < Z < Z_{\alpha/2}) = 1-\alpha$.